Trading More Storage for Less Computation – A KVCache-centric Architecture for Serving LLM Chatbot

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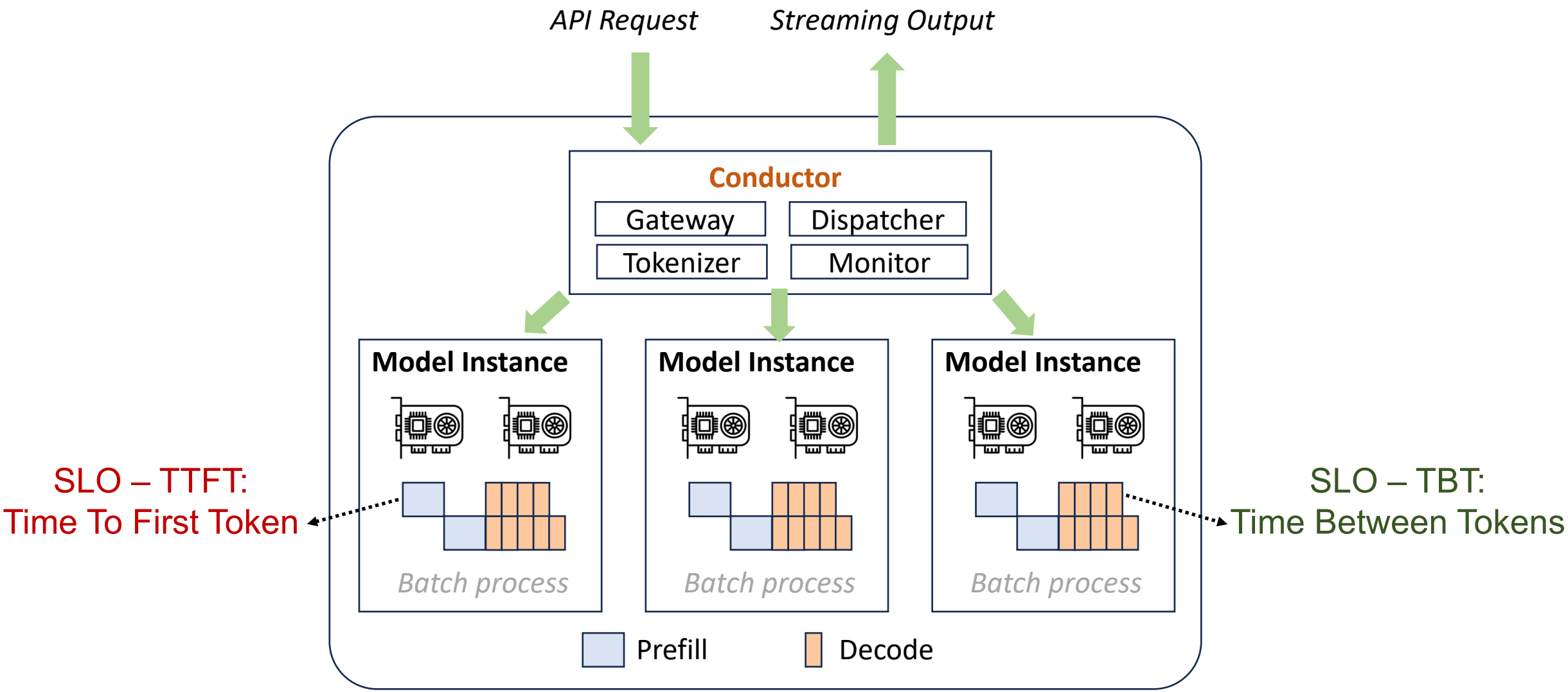
Mingxing Zhang, Yongwei Wu, Weimin Zheng, Xinran Xu

†Co-first Authors

FAST'25

Online LLM Serving System



Online LLM Serving System

Kimi: a leading AI service provided by **Moonshot AI**

- Process over 100 billion tokens daily
- Real-time response
- Up to **1M tokens** context

More Data + Larger Model + Longer Context = ☺ ***Higher Intelligence***

Online LLM Serving System

More Data + Larger Model + Longer Context = ***Higher Intelligence***

BUT

Lack of GPU Supply

Higher Inference Cost

Longer Response Time

Online LLM Serving System

More Data + Larger Model + Longer Context = ***Higher Intelligence***

Reduce the cost for large-scale long-context LLM inference!

Lack of GPU Supply

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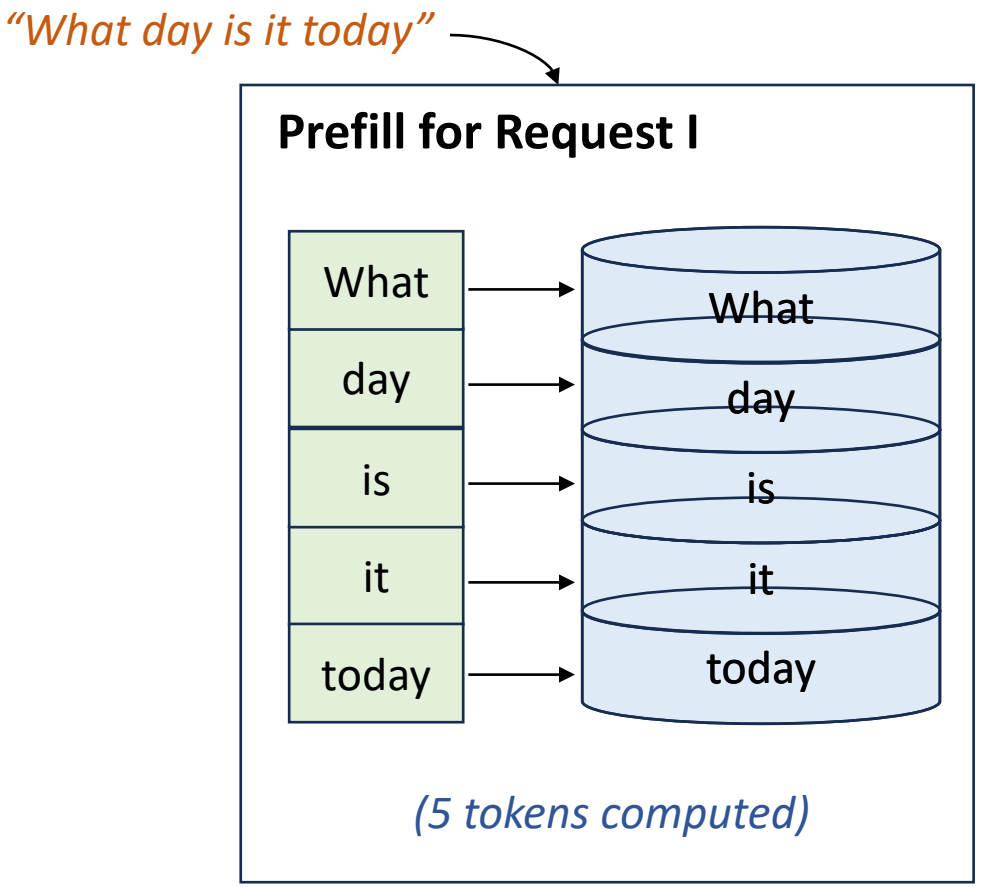
Longer Response Time

LLM Inference: Prefix Caching

- KVCache can be shared across requests with the same prefix, reducing computation during prefill



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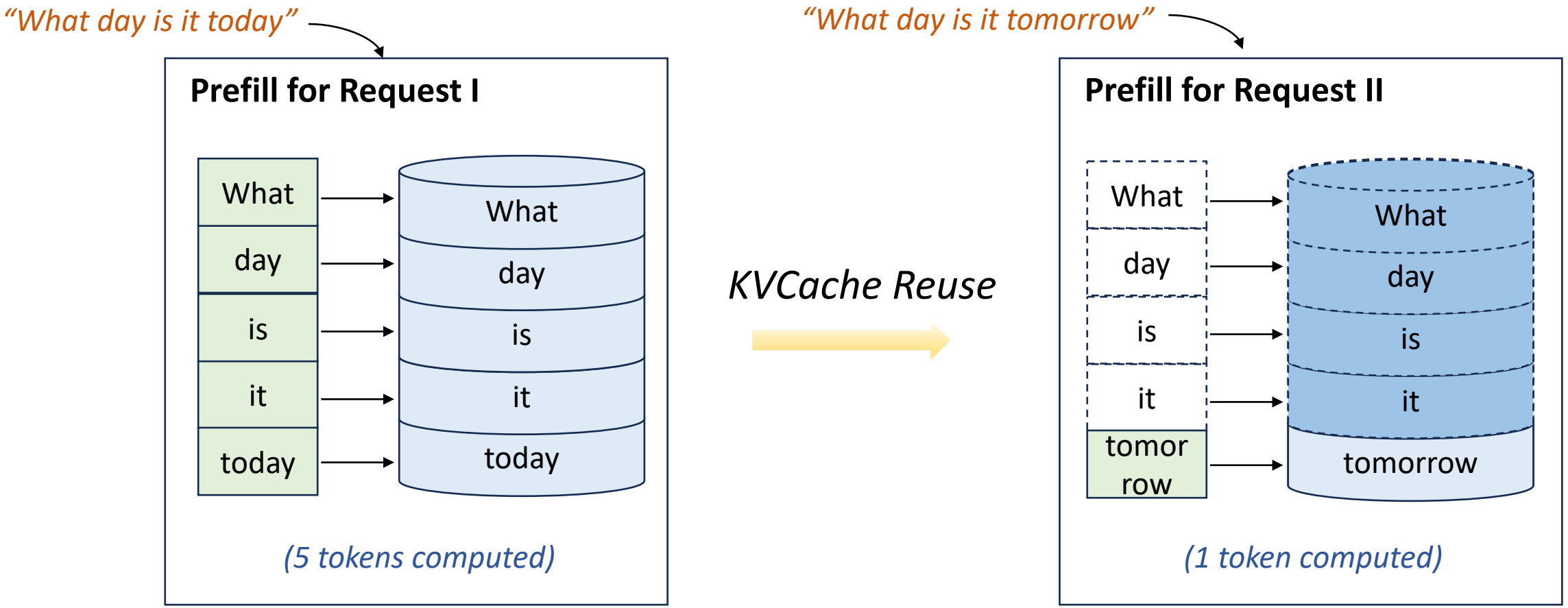
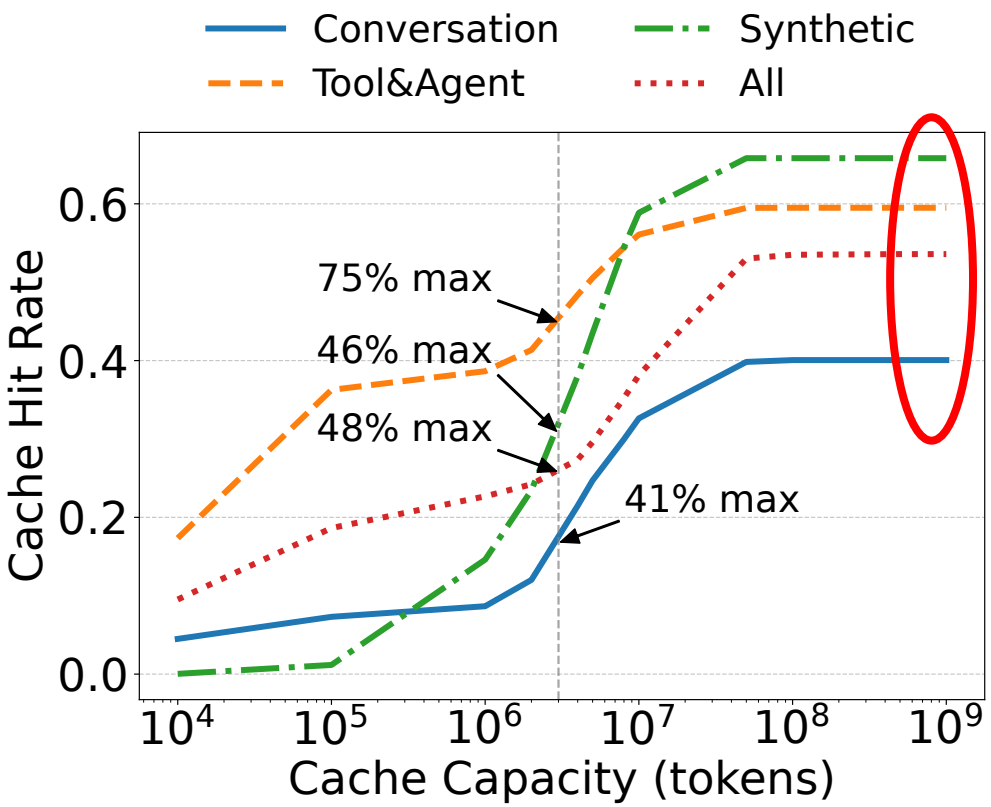


Trace Analysis: Is Prefix Caching a Simple Solution?



Traces in our paper
(open-sourced in https://github.com/kvcache-ai/Mooncake)

Conversation: collected from real-world online conversation requests

Tool&Agent: collected from real-world online requests that include tool use

Synthetic: synthesized from publicly available long context datasets

- Around **50%** of the tokens' KVCache in the real-world workloads can be reused

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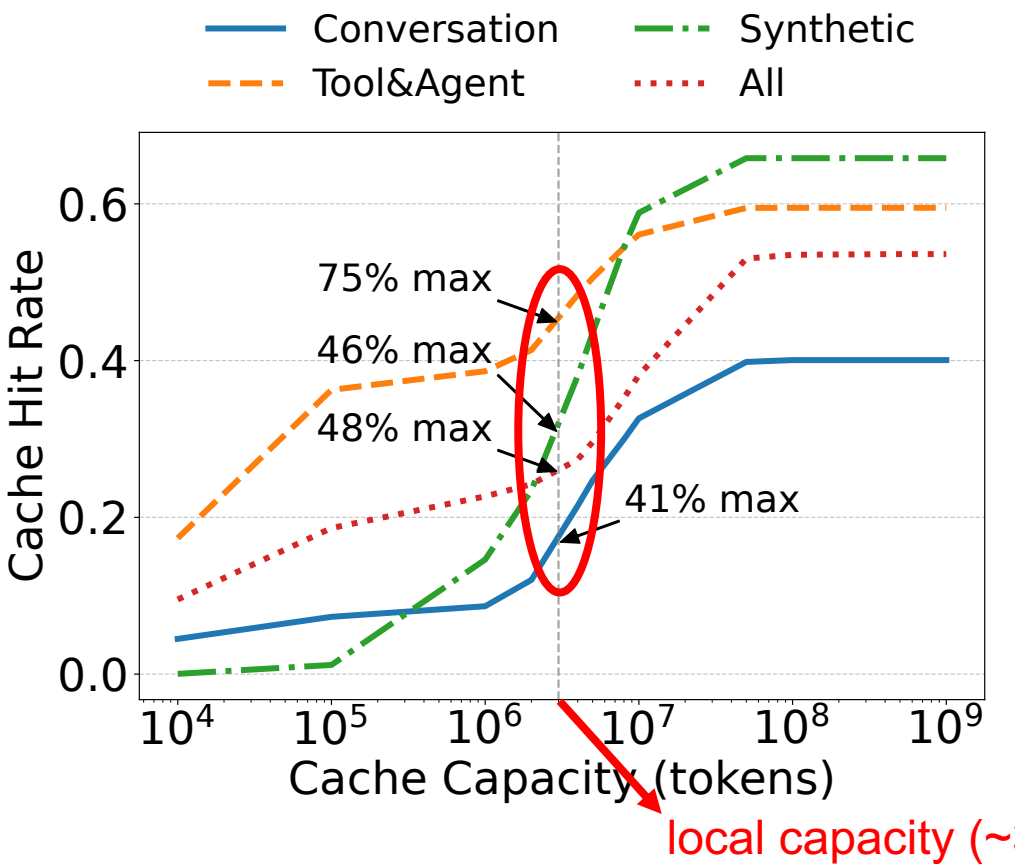

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- Around **50%** of the tokens' KVCache in the real-world workloads can be reused
- **However**, the cache hit rate will significantly drop if only using the local cache

KVCache: Huge Challenge to Storage System

- The volume of reusable KVCache is much larger than the available storage capacity of a single inference node

Over **100 billion** tokens per day

One Token (Bytes)

One Token's KVCache (Tens of KB)

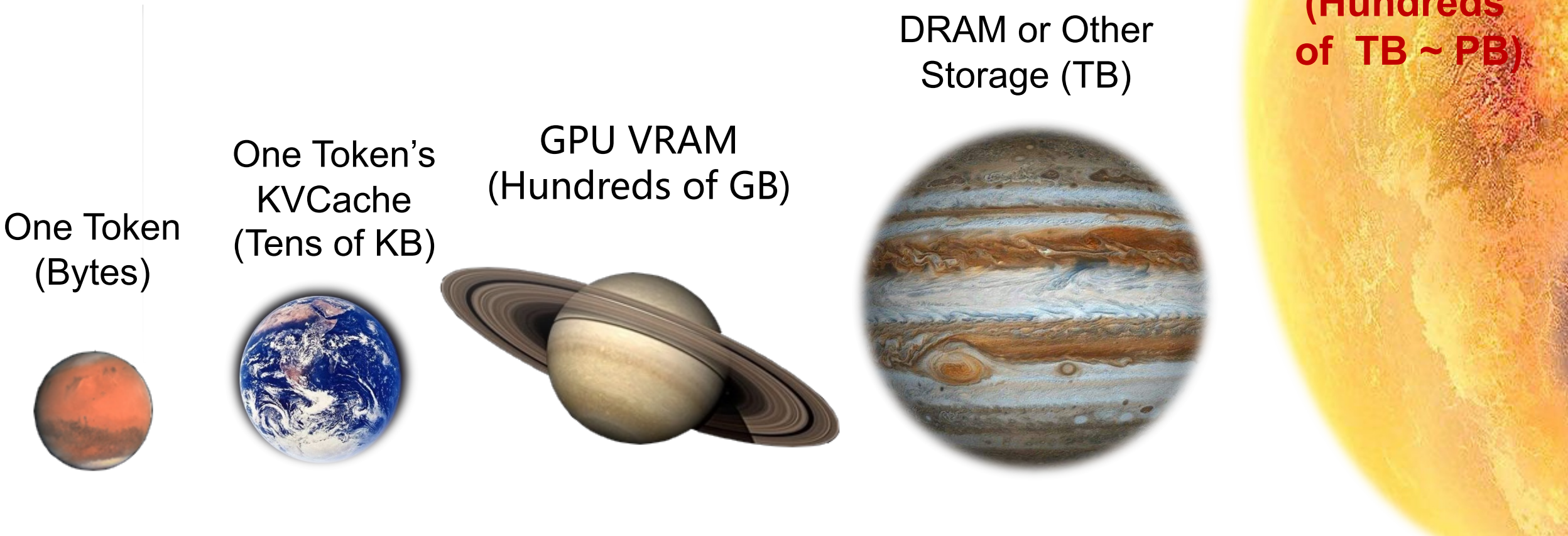
GPU VRAM (Hundreds of GB)

DRAM or Other Storage (TB)

Reusable KVCache (Hundreds of TB ~ PB)

KVCache: Huge Challenge to Storage System

- High transfer bandwidth required to avoid stall of GPU
  - 6 GB/s for A800 GPUs
  - 19 GB/s for H800 GPUs



Trading More Storage for Less Computation

Requirement 1

Large-capacity KVCache storage

Requirement 2

Low-latency and high-bandwidth KVCache transfer

Requirement 3

KVCache-aware scheduling

Trading More Storage for Less Computation

Requirement 1

High-capacity KVCache storage

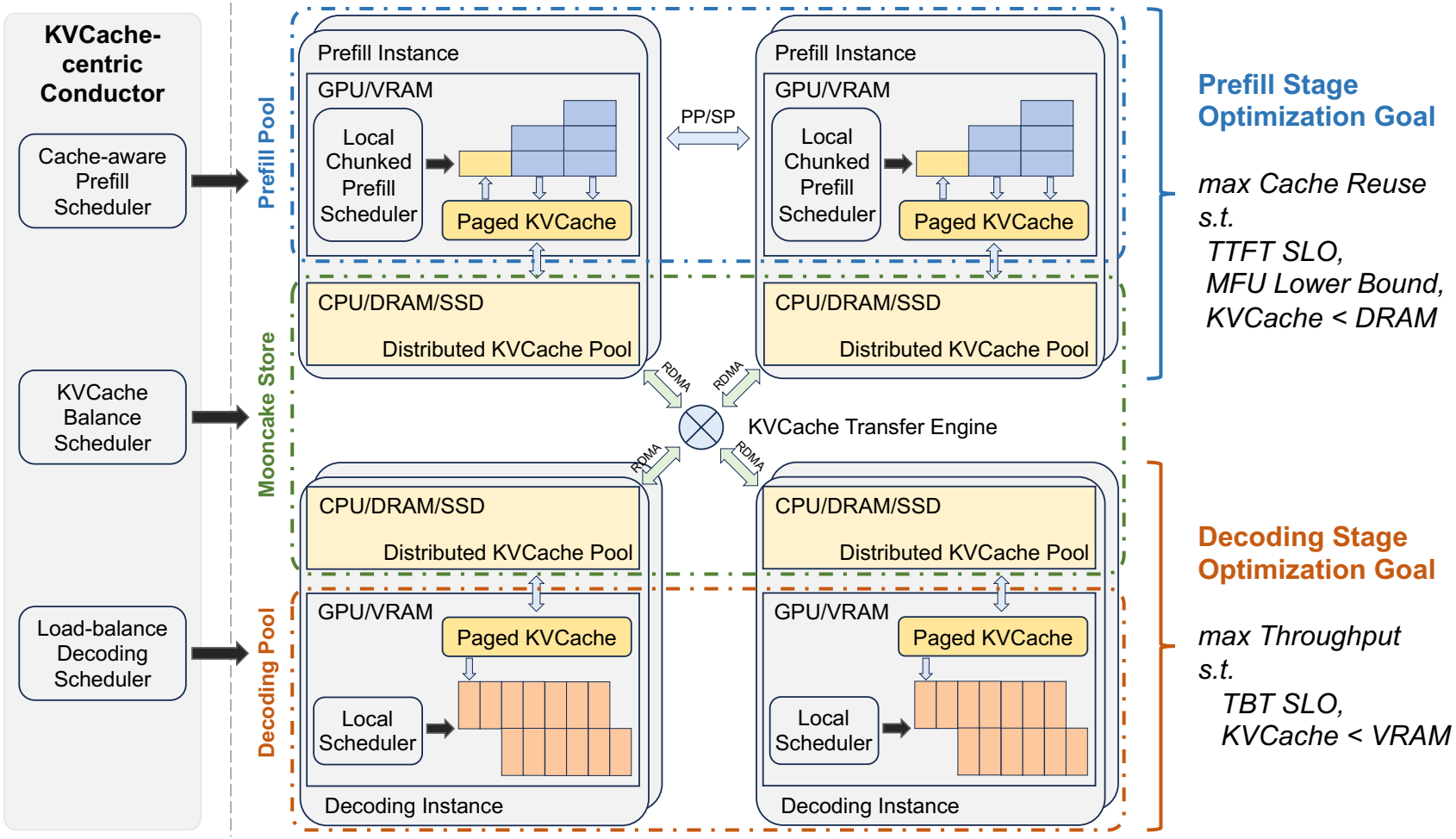
KVCache-centric Architecture is All You Need!

Requirement 3

KVCache-aware scheduling

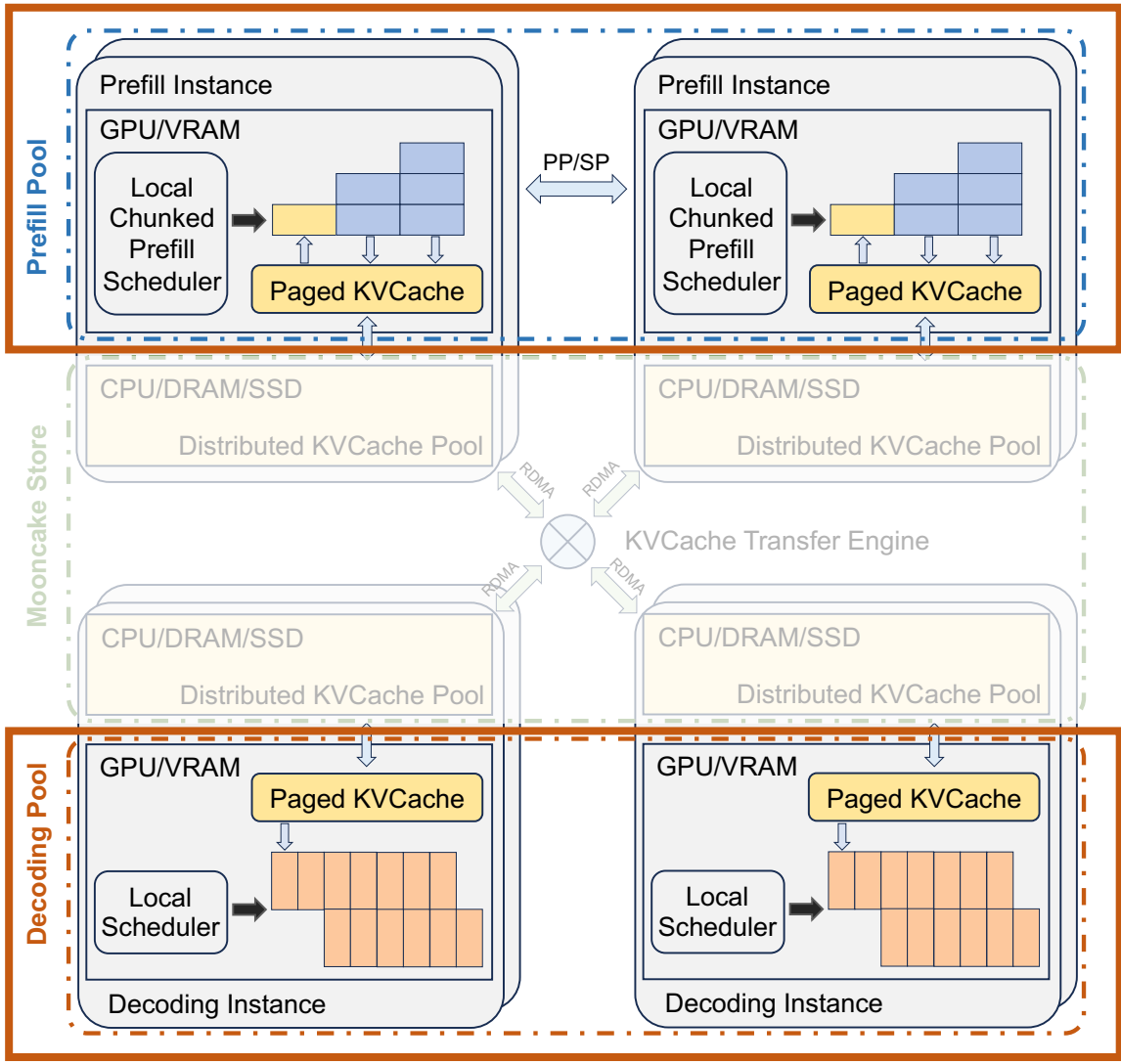
Mooncake: A KVCache-centric Disaggregated Architecture

- P&D disaggregation architecture centered around the distributed KVCache pool – Mooncake Store



Mooncake Store – Distributed Multi-layer KVCache Pool

- Large size and bandwidth (**Key of KVCache Storage**)
  - Distributed KVCache pool across all inference nodes
  - Utilize high performance connection like (GPUDirect) RDMA/Storage
  - Optimized for multi-NIC scenarios
- Design features
  - High scalability and fault tolerance
  - Independent to specific inference engine
  - Flexible API
- Performance
  - Achieve up to a 498% increase in effective request capacity
  - Save 29% - 61% on GPU computation costs



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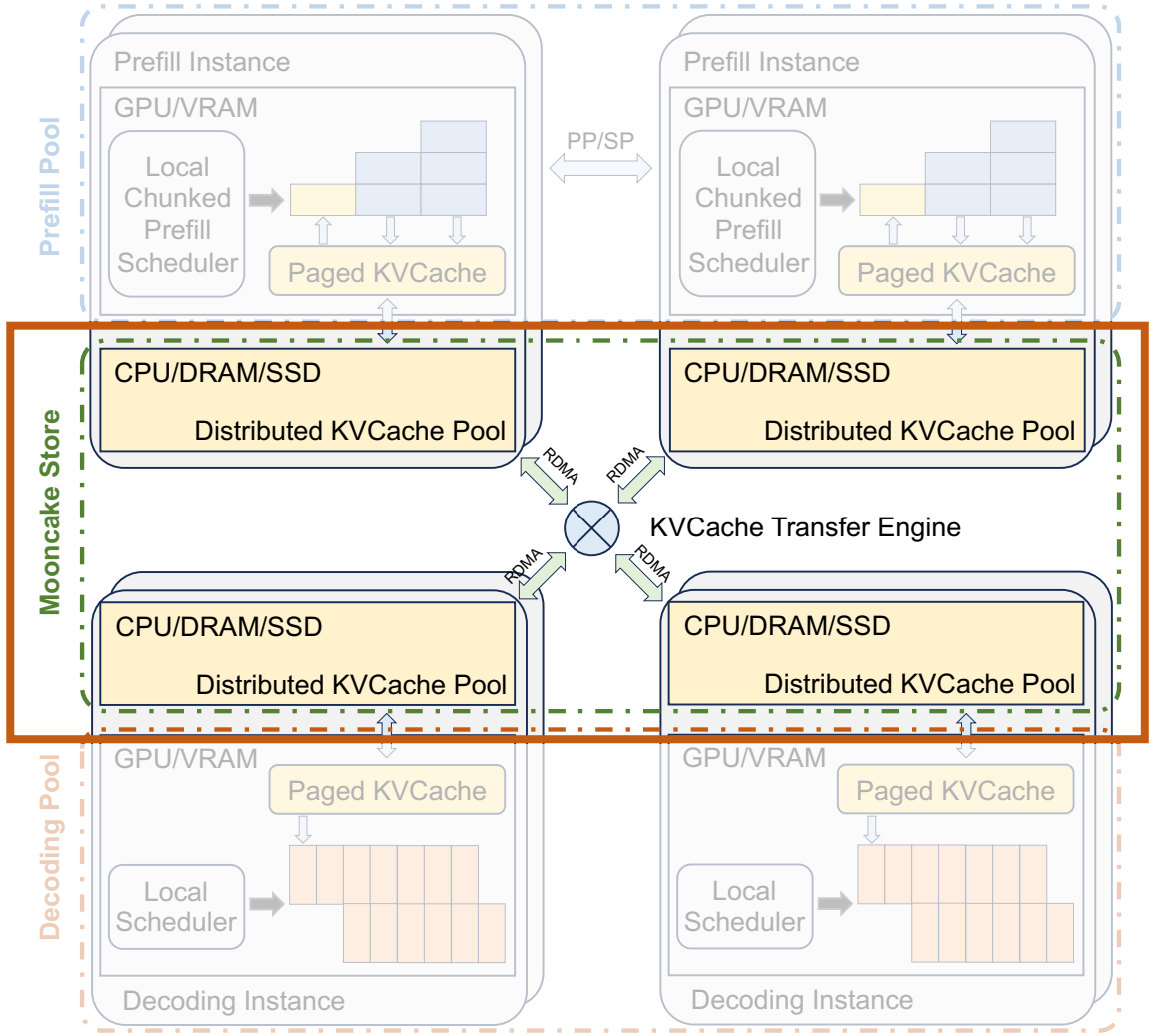
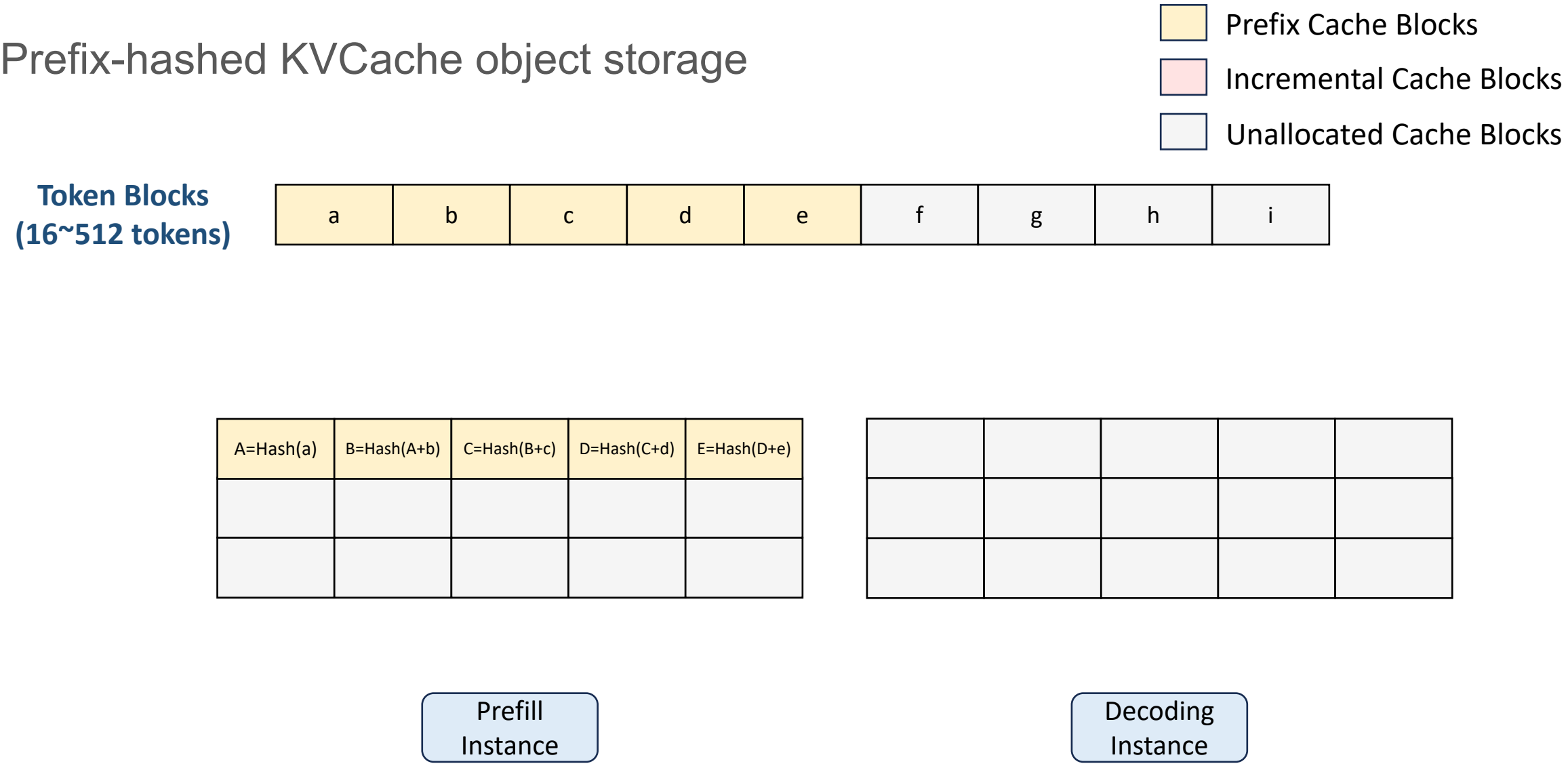


P&D Disaggregation and KVCache Management

- Prefix-hashed KVCache object storage



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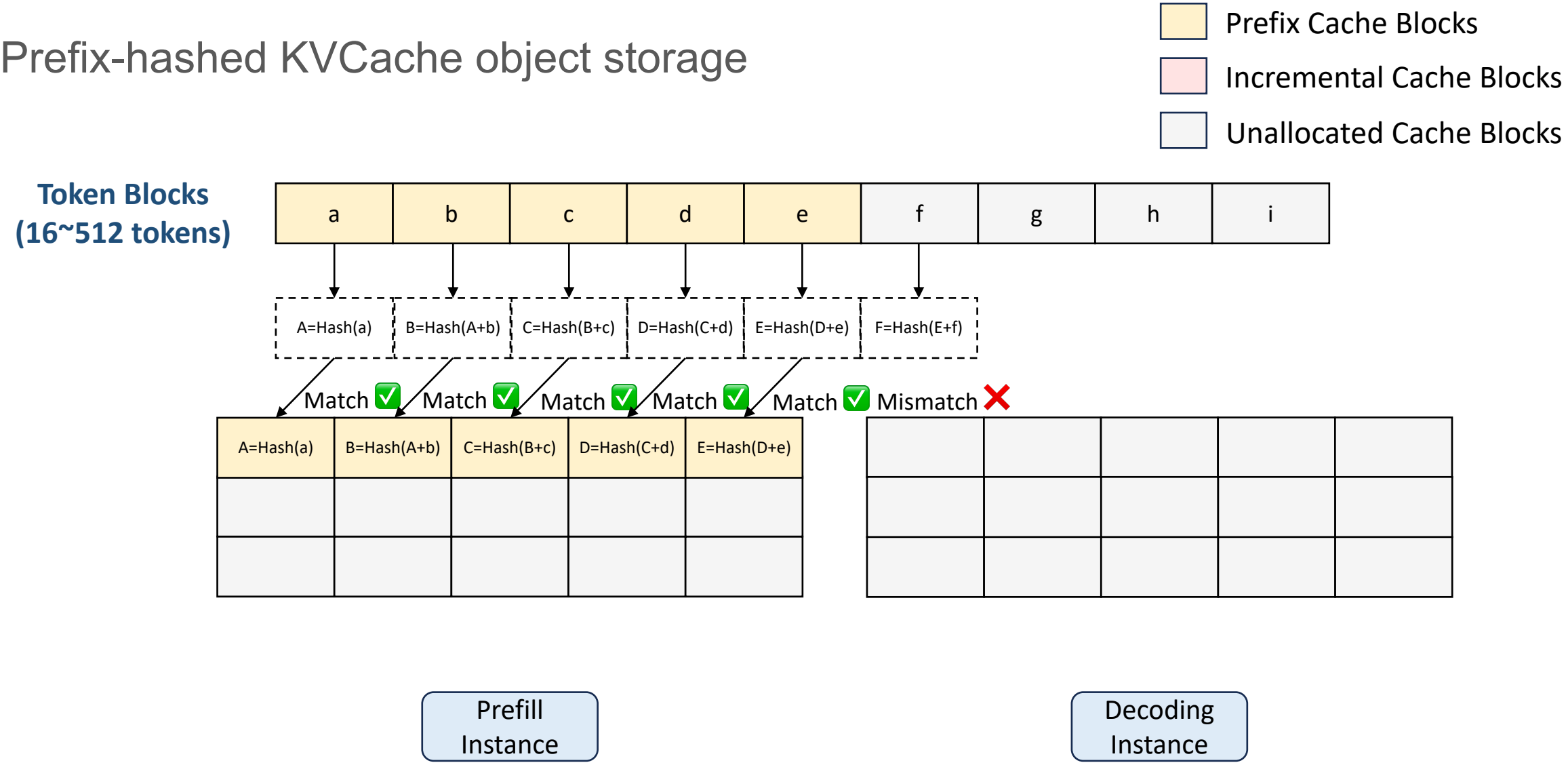

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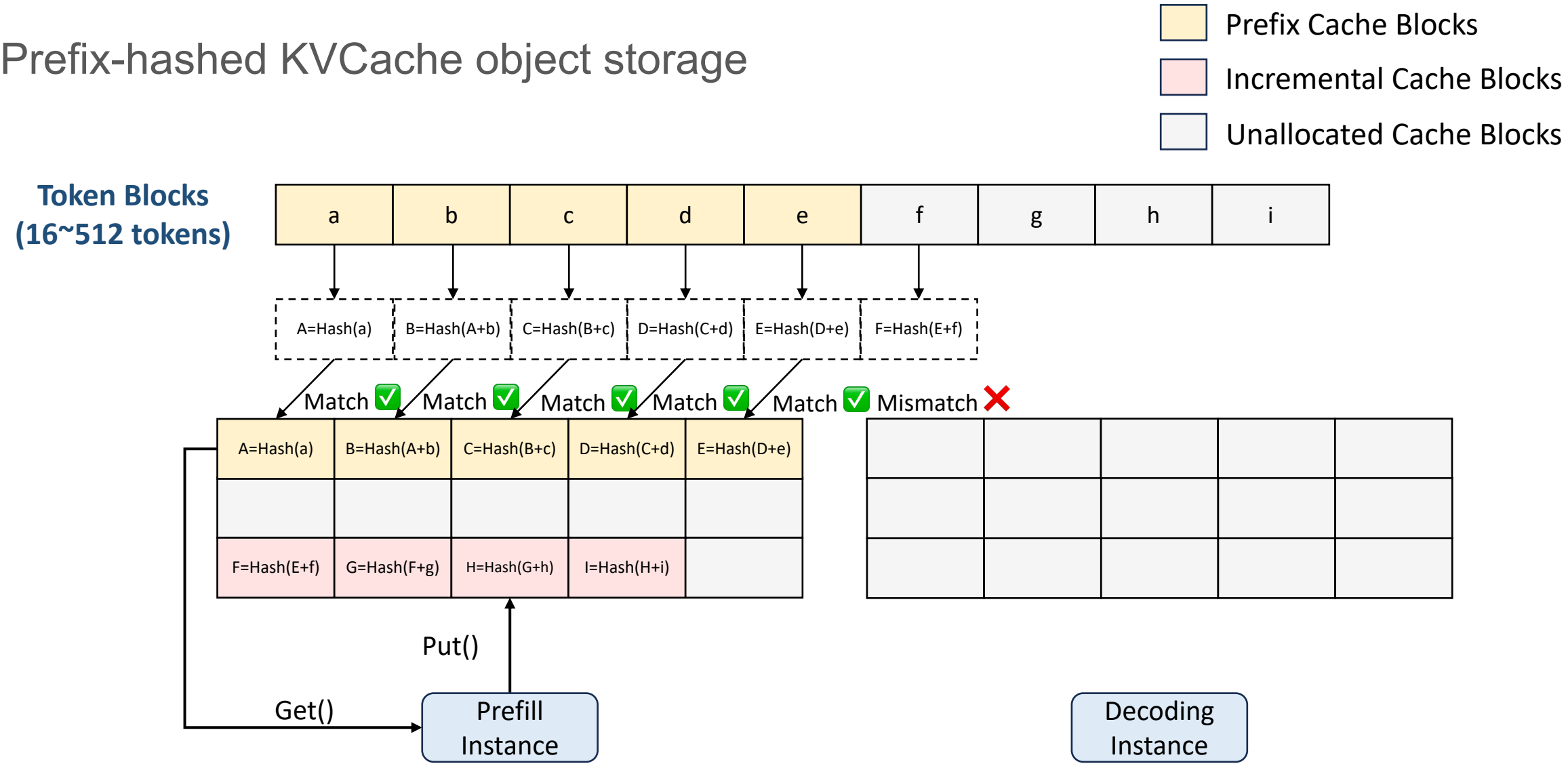

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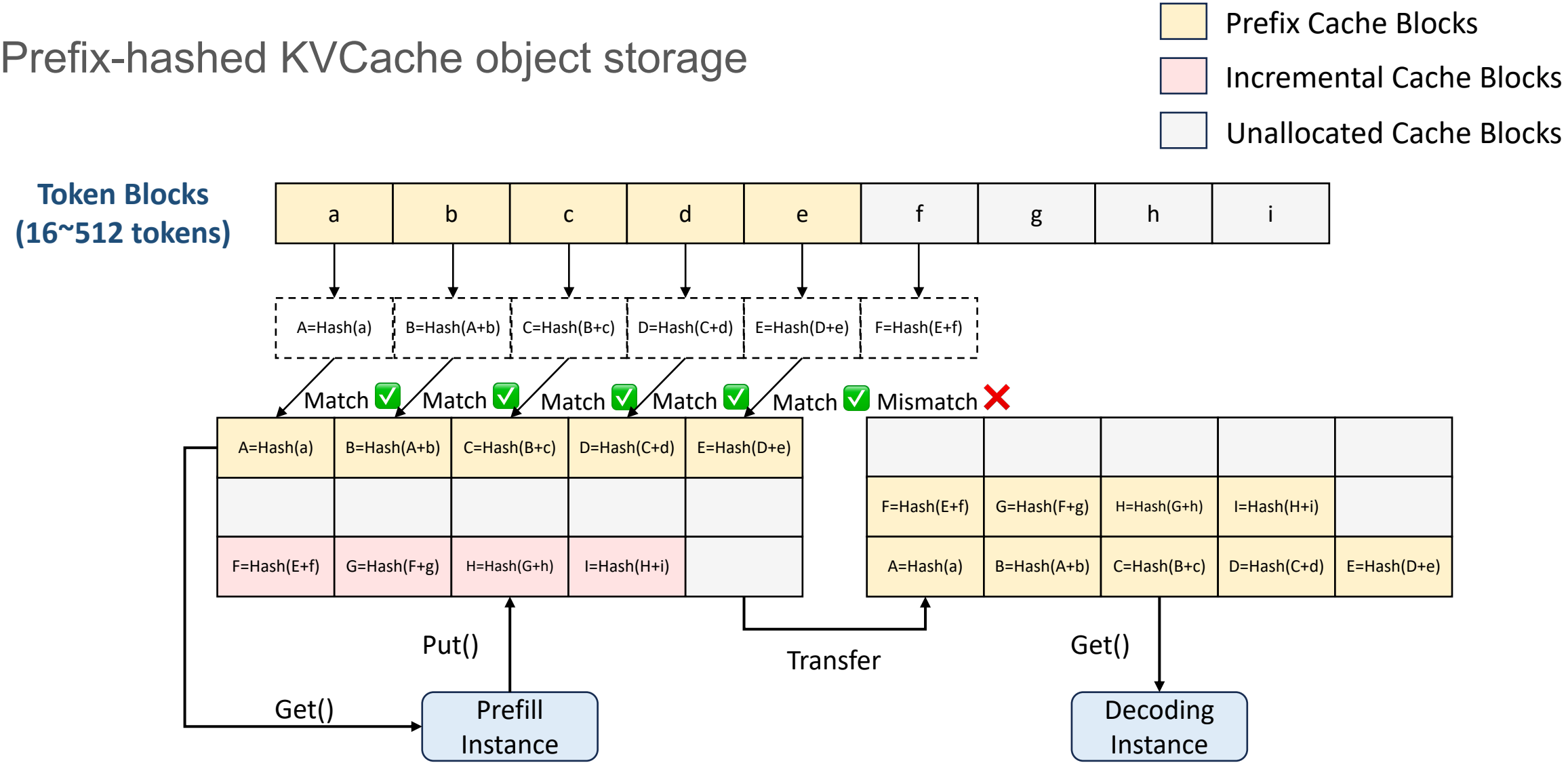
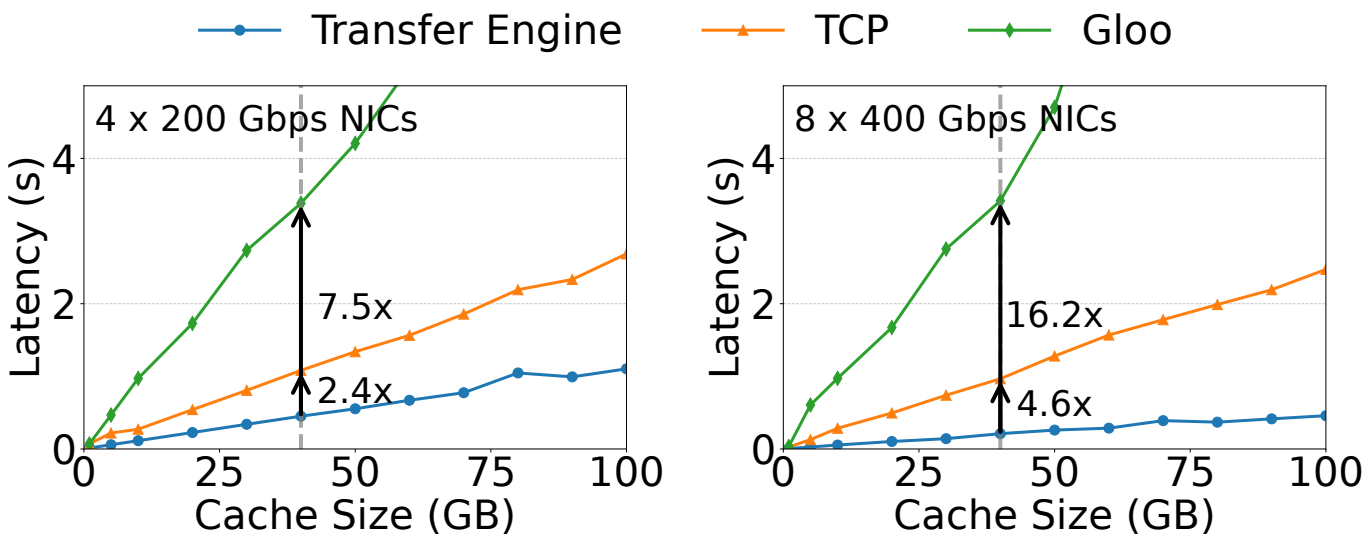
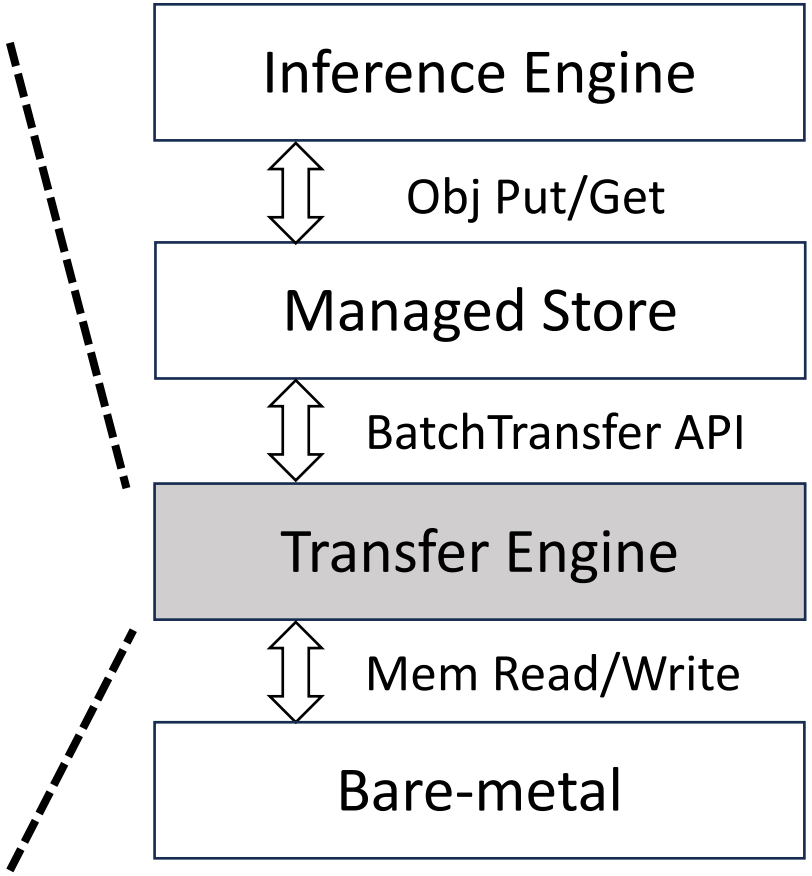


Mooncake Store: Transfer Engine

- **Key technologies**： Zero-copy, multi-NIC up to 8*400Gbps aggregated bandwidth, topology aware, failure recovery, load balance, multi-transport support, etc

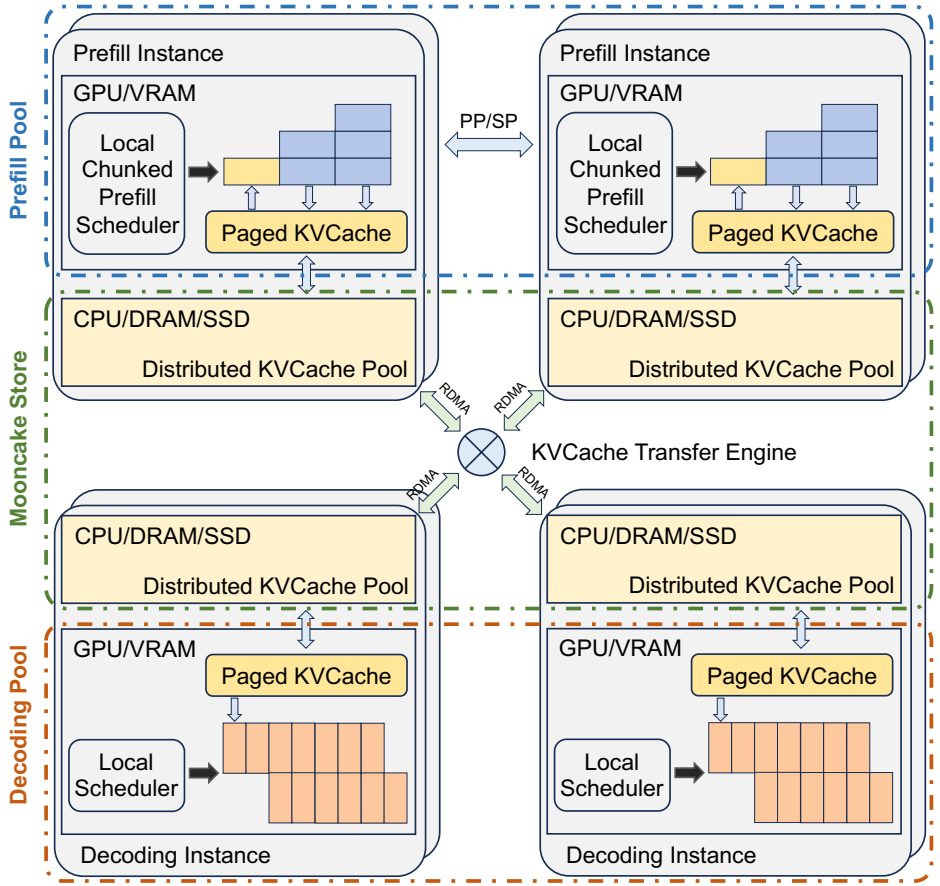


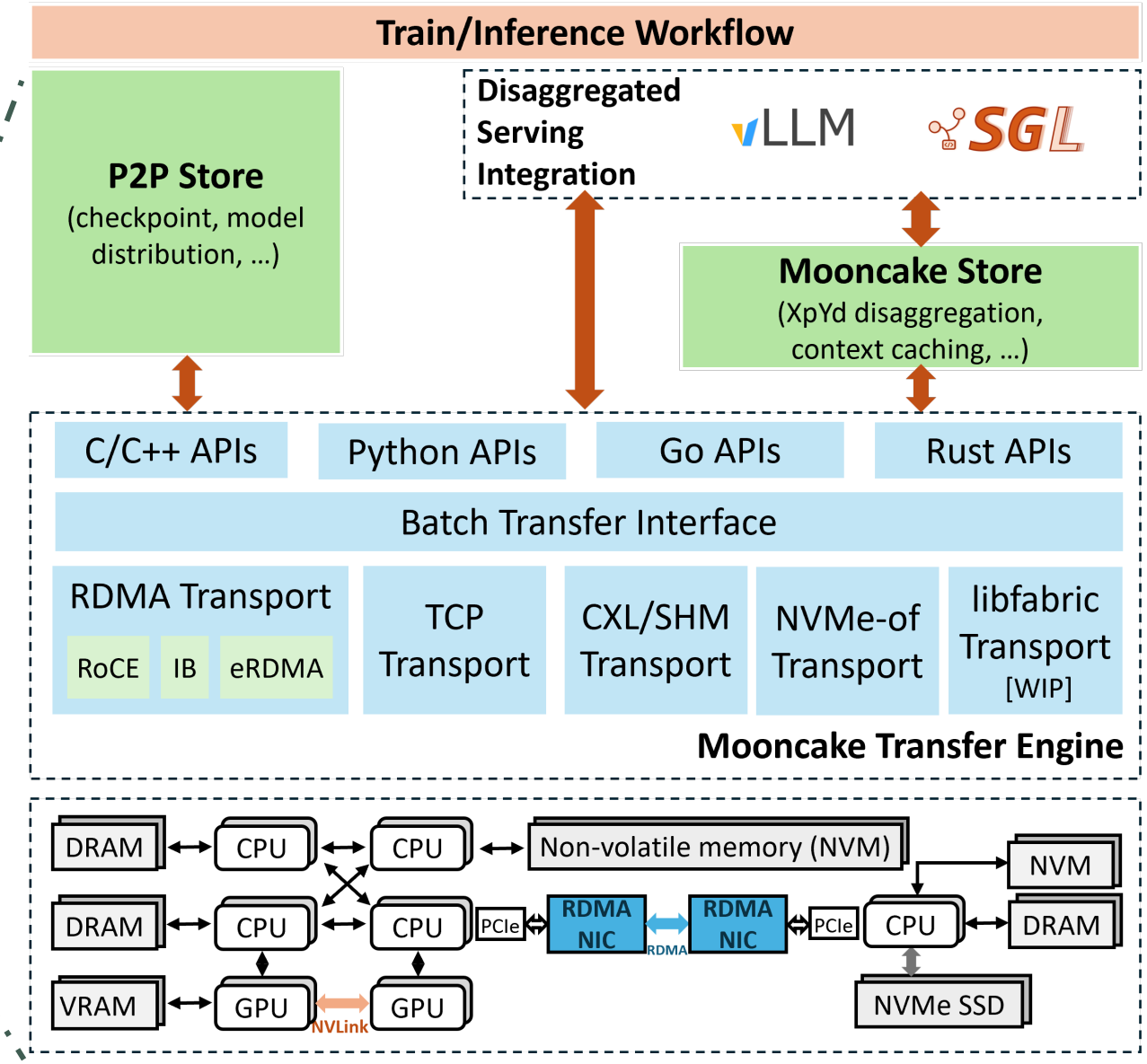


- Better leverage multiple high-performance NICs
- More flexible without building process groups

The Powerful Middleware of Distributed LLM Frameworks

- Mooncake Store is **open-source** and supports a variety of LLM scenarios, including PD disaggregation, KVCache reuse, and RL checkpoint distribution.



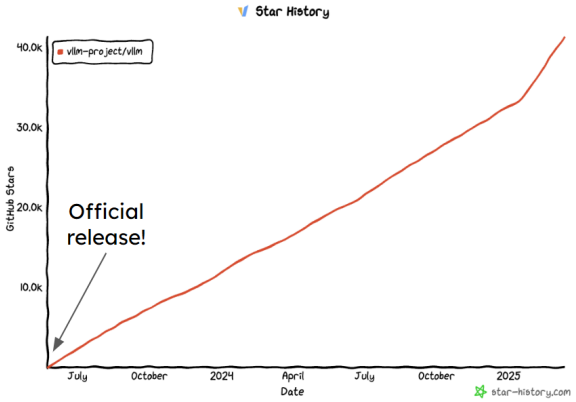


The Open Source Community

vLLM

- **vLLM** - The most widely used inference engine.
- Its distributed inference implementation is built on **Mooncake**.

41.5K Stars



v0.8.3 Latest

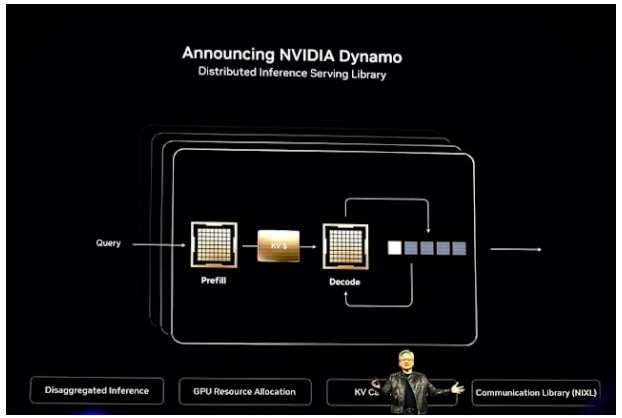
github-actions released this 2 days ago · 57 commits to main since this release v0.8.3 296c657

Cluster Scale Serving

- Support XpYd disaggregated prefill with MooncakeStore (#12957)

NVIDIA Dynamo

- **Dynamo** – the distributed inference system highlighted in the **GTC 2025 Keynote**.
- Its architecture is inspired by **Mooncake**, with explicit acknowledgment in its documentation.



Acknowledgement

We would like to acknowledge several open source software stacks for motivating us to create Dynamo.

- vLLM and vLLM-project
- SGLang
- DistServe
- Mooncake

- *Memory bottlenecks*: Large-scale inference workloads demand extensive capacity. KV cache offloading across memory hierarchies (HBM, DDR, N memory limits and speeds up latency. (Mooncake, AIBrix, LMCache)

SGL

- **SGLang** - the core inference engine of xAI, is widely used in DeepSeek's inference stack.
- Its distributed inference architecture is jointly developed in collaboration with **Mooncake**.

LMSYS Org
@lmsysorg

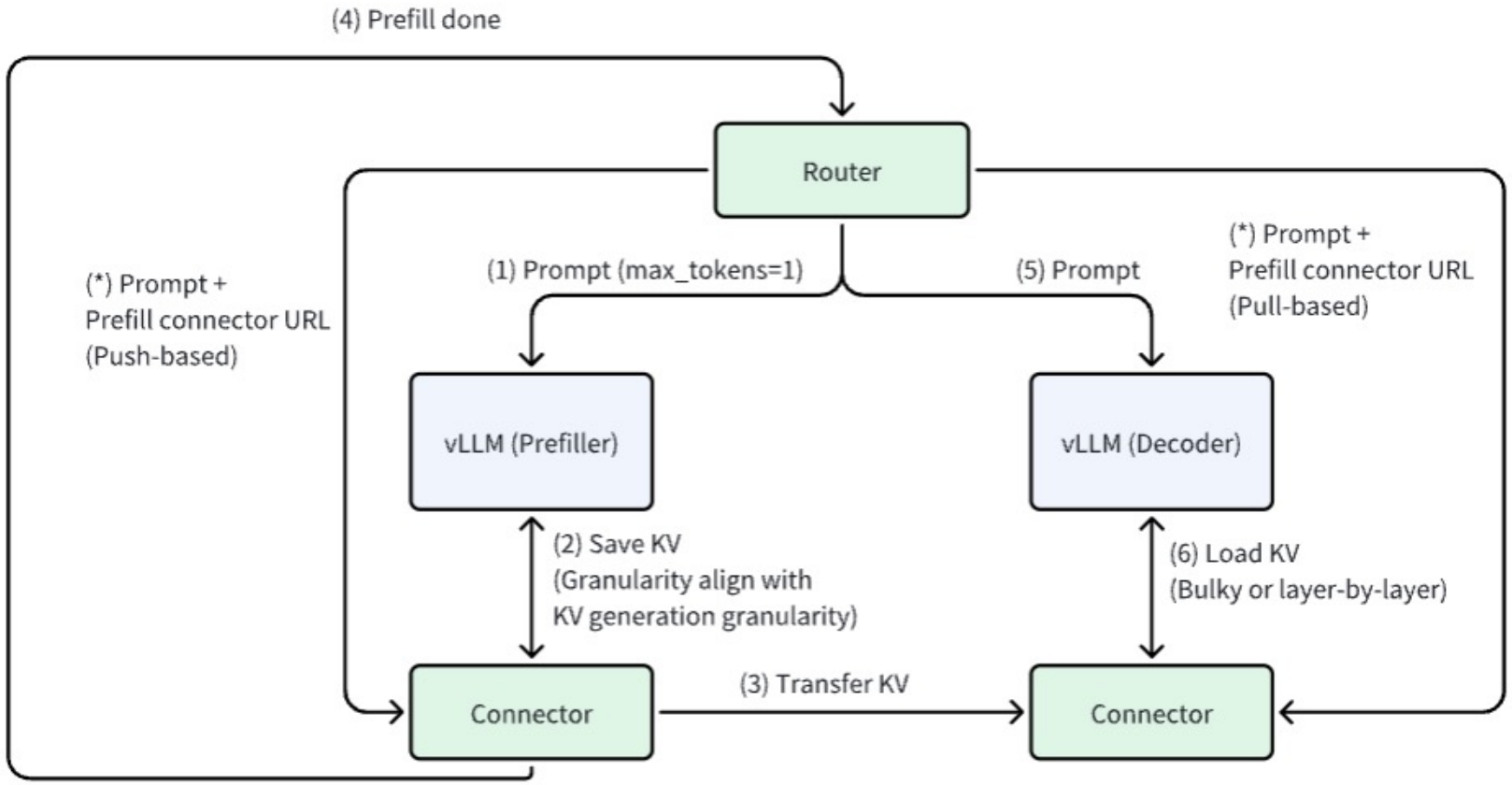
The SGLang Team is honored to announce that the following well-known companies and teams, among others, have adopted SGLang for running DeepSeek V3 and R1. @AMD @nvidia @Azure @basetenco @novita_ai_labs @BytedanceTalk @DataCrunch_io @hyperbolic_labs @Vultr @runPod

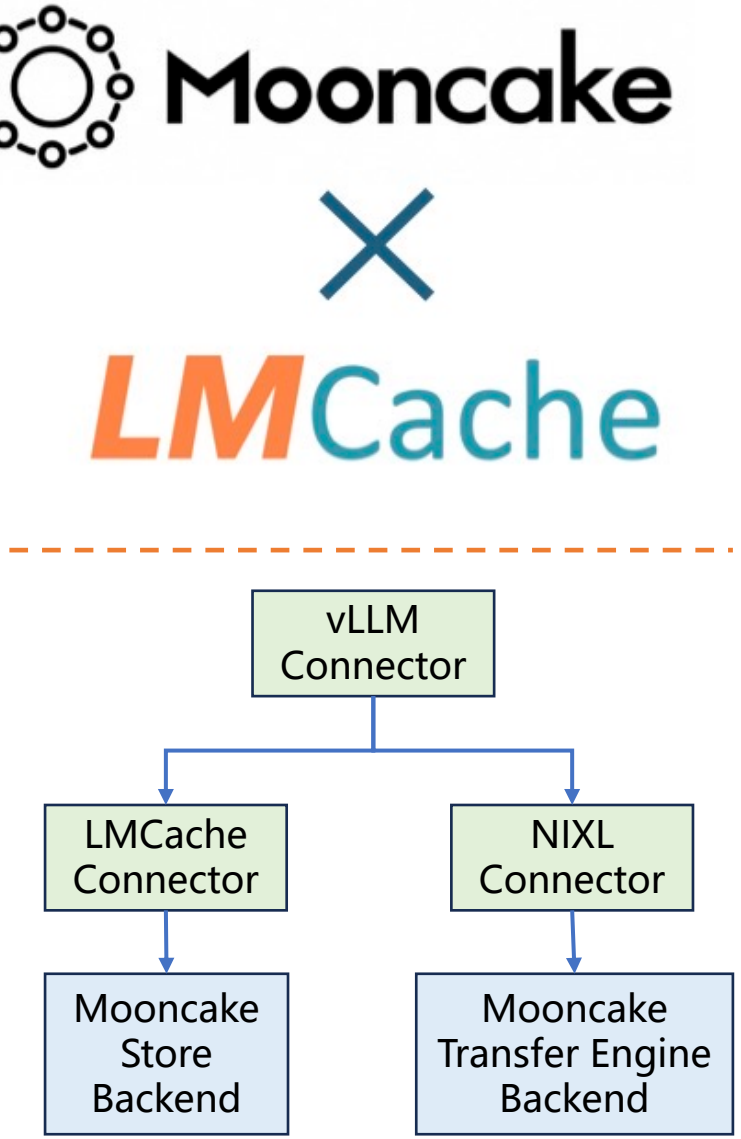
LMSYS Org
@lmsysorg

SGLang has achieved a milestone with full support for PD Disaggregation, thanks to the MoonCake team. Teng Ma, Shangming Cai, Xuchun Shang and Yuan Luo were instrumental in this achievement. Special thanks to Atlas Cloud for their support with the H100s cluster. Let's go! 🚀

Mooncake + vLLM

- Integrated with vLLM as the backend of the LMCache Connector





https://blog.lmcache.ai/2025-05-08-mooncake

Mooncake + SGLang

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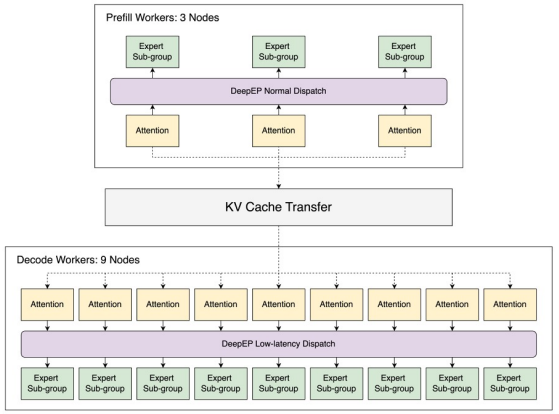
🚀 Breaking: SGLang provides the first open-source implementation to serve @deepseek_ai V3/R1 models with large-scale expert parallelism and prefill-decode disaggregation on 96 GPUs.
It nearly matches the throughput reported by the official DeepSeek blog, achieving 52.3K input tokens per second and 22.3K output tokens per second per node. This optimized strategy improves output throughput by up to 5x compared to vanilla tensor parallelism.



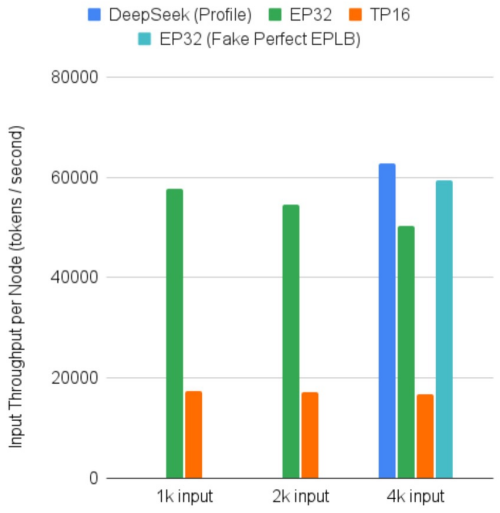
- 3 Prefill+ 9 Decode
- DeepEP + EPLB
- Two-batch Overlap

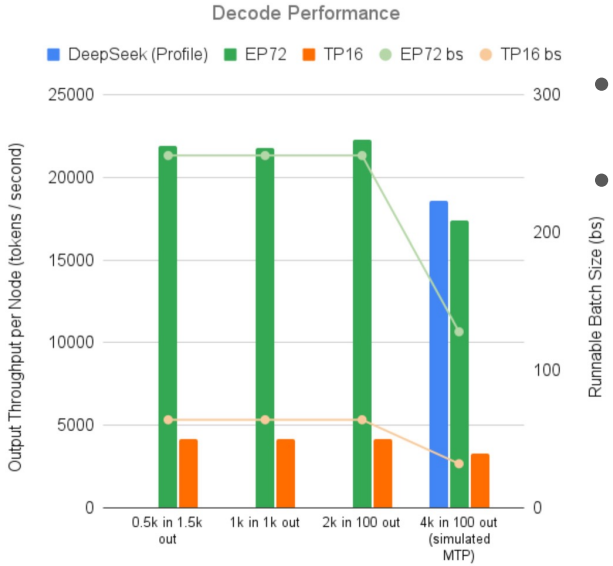
Acknowledgment

We would like to express our heartfelt gratitude to the following teams and collaborators:

- **SGLang Core Team and Community Contributors** — Jingyi Chen, Cheng Wan, Liangsheng Yin, Baizhou Zhang, Ke Bao, Jiexin Liang, Xiaoyu Zhang, Yanbo Yang, Fan Yin, Chao Wang, Laixin Xie, Runkai Tao, Yuhong Guo, Kaihong Zhang, Lei Yu, Yu-Hsuan Tseng, Qilin Tian, Peng Zhang, Yi Zhang, Yineng Zhang, Byron Hsu, and many others.
- Atlas Cloud **Team** — Jerry Tang, Wei Xu, Simon Xue, Harry He, Eva Ma, and colleagues — for providing a 96-device NVIDIA H100 cluster and offering responsive engineering support.
- **NVIDIA Solution Architect Team** — Xuting Zhou, Jinyan Chen, and colleagues — for their work on the seamless integration of expert parallelism.
- **NVIDIA Enterprise Product Team** — Trevor Morris, Elfie Guo, Kaixi Hou, Kushan Ahmadian, and colleagues — for optimizing the DeepSeek R1 kernels.
- **LinkedIn Team** — Biao He, Qingquan Song, Chunan Zeng, Yun Dai, Yubo Wang, and colleagues — for optimizing the Flash-Attention 3 backend.
- **Mooncake Team** — Shangming Cai, Teng Ma, Mingxing Zhang, and colleagues — for their collaboration on PD disaggregation in SGLang.
- **FlashInfer Team** — Zihao Ye, Yong Wu, Yaxing Cai — for additional DeepSeek R1 kernel optimizations.
- **Dynamo Team** - Kyle Kranen, Vikram Sharma Mailthody, and colleagues - for extra support on PD disaggregation in SGLang.

Thank you all for your invaluable support and collaboration.





- 52.3k input tps
- 22.3k output tps

https://lmsys.org/blog/2025-05-05-large-scale-ep

Mooncake + SGLang

NVIDIA AI Developer
@NVIDIAAIDev

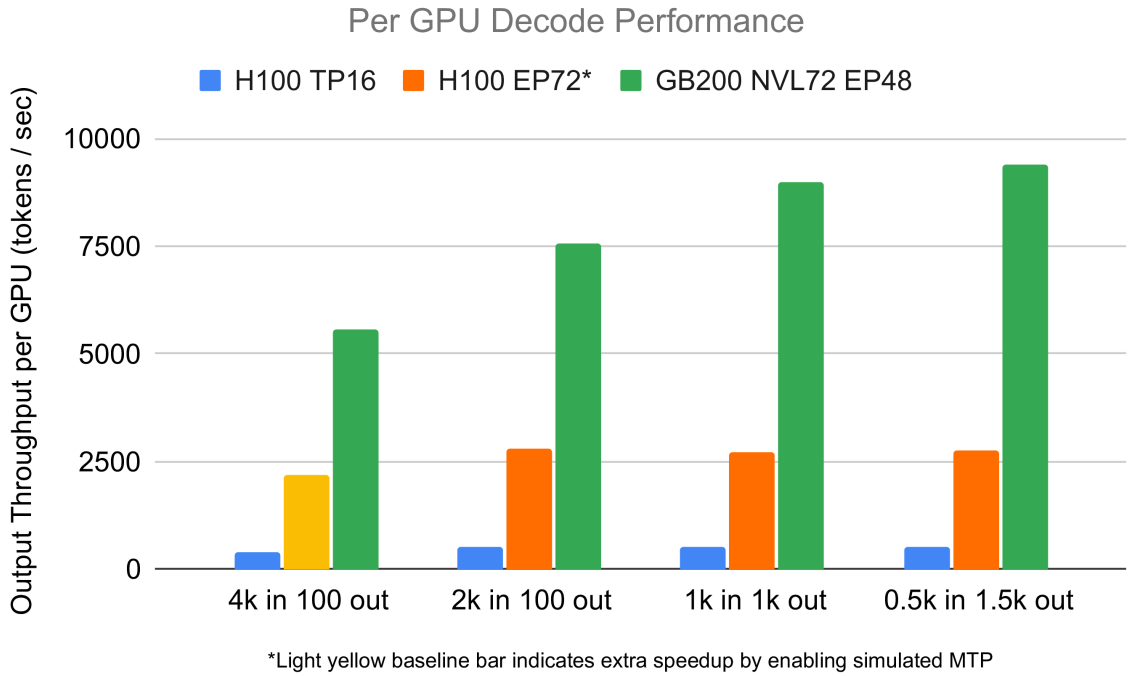
.@lmsysorg (SGLang) now achieves 7,583 tokens per second per GPU running @deepseek_ai R1 on the GB200 NVL72, a 2.7x leap over H100.

We're excited to see the open source ecosystem advance inference optimizations on GB200 NVL72, driving down cost per token for the industry at large.

This leap was made possible by SGLang's open source, Blackwell-specific software optimizations, which harness the rack-scale GB200 NVL72's scale-up capabilities and ultra-low-latency GPU-to-GPU interconnects. This progress was enabled through close collaboration.

👏 Huge thanks to:
- #SGLang Core Team & Community
- #FlashInfer Team
- Mooncake Team
- NVIDIA Teams from Hardware, DevTech, DGX Cloud, Dynamo, and Enterprise Product — for powering GB200 NVL72 ops and Blackwell DeepGEMM breakthroughs

Deploying DeepSeek on GB200 NVL72 with PD and Large Scale EP (Part I): 2.7x Higher Decoding Throughput



In collaboration with SGLang and NVIDIA, we are the first to support the NVL72 supernode, achieving a 2.7× performance improvement over H100.

https://lmsys.org/blog/2025-06-16-gb200-part-1

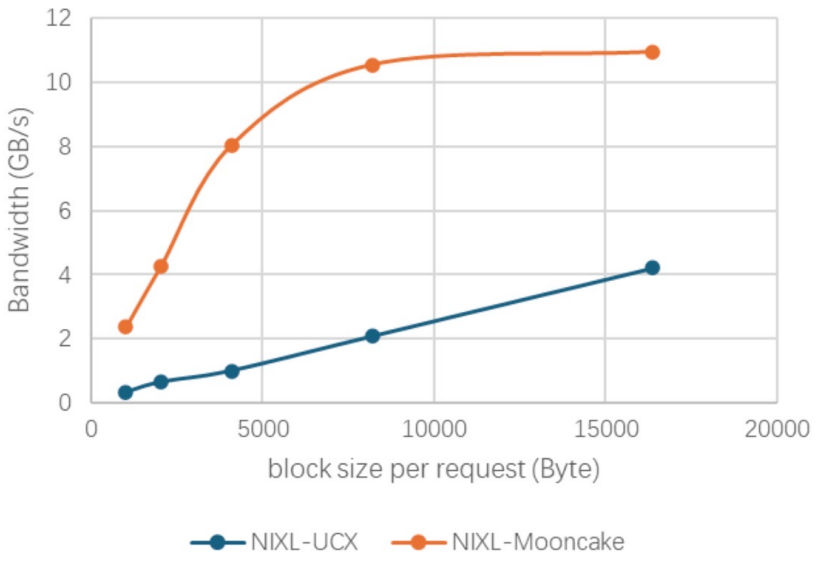
Mooncake + Dynamo

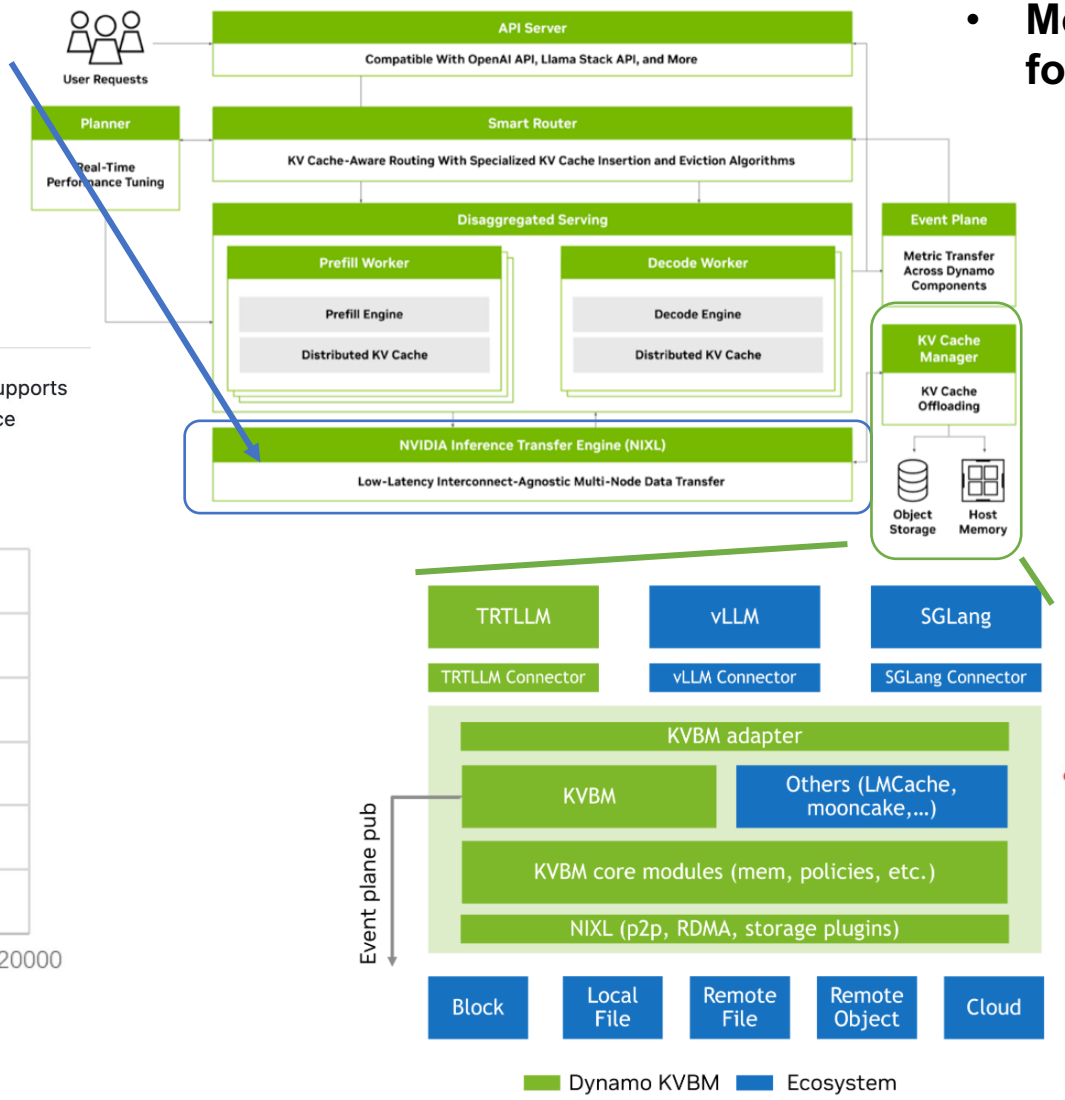
- **Mooncake Transfer Engine as the Backend for NIXL**

  Demonstrates significant performance advantages in small chunk and multi-NIC scenarios

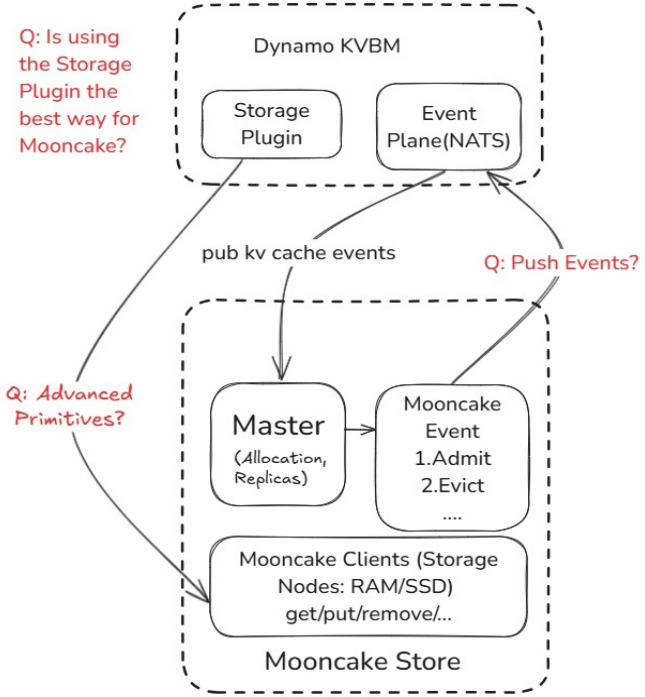
Performance Metric

We use two machines in the same 200Gbps InfiniBand network (due to the CPU supports only PCIe 3.0, the maximal bandwidth is about 11 GB/s) to evaluate the performance between UCX backend and Mooncake Backend. The test program is `examples/cpp/nixl_benchmark.cpp` and we only change the `block_size` param.





- **Mooncake Store as the Backend for KVBM**



Join the Open Source Community for Efficient LLM Serving

kvcache.ai

KVCache.AI is a joint research project between MADSys and top industry collaborators, focusing on efficient LLM serving.

686 followers https://madsys.cs.tsinghua.edu.cn/ zhang_mingxing@mail.tsinghua.edu.cn

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Mooncake Public

Mooncake is the serving platform for Kimi, a leading LLM service provided by Moonshot AI.

C++ 3k 194

ktransformers Public

A Flexible Framework for Experiencing Cutting-edge LLM Inference Optimizations

Python 13.4k 925

https://github.com/kvcache-ai

Summary

Deployed System: Mooncake

- Trading more storage for less computation
- A KVCache-centric LLM serving architecture for higher throughput

Infrastructure: Mooncake Store

- Fast, scalable and fault-tolerant KVCache storage and transfer engine
- Open-sourced with support for multiple LLM serving frameworks

Benefits of Mooncake Store

- Efficient P&D disaggregation
- More KVCache reuse
- Flexible KVCache scheduling

Paper

GitHub